$36°$. Thus a regular 10-gon which has central angles of $36°$ is constructible. A regular pentagon is obtain by starting at vertex 1 of a regular 10-gon and drawing line segments to vertex 3, on to vertex 5, on to vertex 7, on to vertex 9, and then to vertex 1.

6. A regular 20-gon is constructible because we can bisect the constructible angle of $36°$ (see Exercise 5) to obtain an angle of $18° = (360/20)°$.

7. Because we can construct an angle of $72° = 2(36°)$ by Exercise 5, and because $60°$ is a constructible angle, we can construct an angle of $72° - 60° = 12° = (360/30)°$. Therefore a regular 30-gon can be constructed.

8. Exercise 7 shows that a $12°$ angle can be constructed, so a $24° = (2 \cdot 12)° = (72/3)°$ angle can be constructed. Thus an angle of $72°$ can be trisected.

9. Exercise 7 shows that an angle of $12°$ can be constructed so an angle of $24° = (2 \cdot 12)° = (360/15)°$ can be constructed. Hence a regular 15-gon can be constructed.

10. Starting with a line segment of length 1, from 0 to 1 on the usual x-axis, we can construct a line segment of any rational length in a finite number of steps, and thus find any point with rational coordinates in the x, y-plane in a finite number of steps. Thinking analytically, the only other points in the plane we can locate must appear as an intersection of two lines, of a line and a circle, or of two circles, which reduces algebraically to finding the solutions of only linear or quadratic equations. Considering a right triangle with an acute angle θ and hypotenuse of length 1, we see that we can construct the angle θ if and only if we can construct a line segment of length $\cos\theta$. It can be shown that $\cos 20°$ is not a solution of a linear or quadratic equation, but rather of a cubic equation, so while we can construct a $60°$ angle because $\cos 60° = 1/2$, we cannot trisect it to obtain a $20°$ angle.

33. Finite Fields

1. Because $4096 = 2^{12}$ is a power of a prime, a finite field of order 4096 does exist.

2. Because $3127 = 53 \cdot 59$ is not a power of a prime, no finite field of order 3127 exists.

3. Because $68921 = 41^3$ is a power of a prime, a finite field of order 68921 does exist.

4. $GF(9)^*$ is a cyclic group under multiplication of order 8 and has $\phi(8) = 4$ generators, so there are 4 primitive 8th roots of unity.

5. $GF(19)^*$ is a cyclic group under multiplication of order 18 and has $\phi(18) = 6$ generators, so there are 6 primitive 18th roots of unity.

6. $GF(31)^*$ is a cyclic group under multiplication of order 30. Its cyclic subgroup of order 15 has $\phi(15) = 8$ generators, so it contains 8 primitive 15th roots of unity.

7. $GF(23)^*$ is a cyclic group under multiplication of order 22. Because 10 is not a divisor of 22, it contains no elements of order 10, so $GF(23)^*$ contains no primitive 10th roots of unity.

8. T F T F T F T T F T